

Big Data Analysis for the NBA Players

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Team Name: Full Court Insights

1. Introduction and Motivation

The NBA player performance analytics domain presents a compelling case for big data analysis. Professional basketball generates massive volumes of granular data, with over 1.35 million individual play-by-play events captured across seasons, each containing detailed information about shots, passes, rebounds, steals, blocks, turnovers, and fouls. This depth of event-level data allows for unprecedented scrutiny of player contribution and impact. NBA MVP voting historically relies on subjective media narratives, team success metrics, and recency bias rather than comprehensive statistical evidence. Our central research question is: Which players deserve MVP recognition based purely on play-by-play data analysis, independent of media bias?

Specifically, we seek to identify:

- Players whose clutch performance elevates them above season-long averages
- Efficient role players whose impact is overlooked by traditional volume-based metrics
- Era-adjusted performance to ensure fair MVP comparison across decades
- Undervalued performers whose contributions are missed by conventional MVP voting

A big data approach is necessary because traditional methods (manual box score analysis, single-metric comparison) cannot handle:

- Scale: 1.35M events × 20+ derived features across 30+ years of NBA data
- Complexity: Nuanced metrics like points per possession, clutch-time efficiency, era-adjusted baselines require distributed processing
- Multidimensionality: Balancing volume vs. efficiency vs. clutch performance vs. defensive impact simultaneously
- Real-time Processing: Future iterations could ingest live game data for predictive MVP rankings

PySpark enables efficient distributed processing of this scale, allowing exploration of relationships that single-machine analysis cannot handle practically. MLlib classification models can predict official MVP vote shares from our engineered features, revealing the hidden statistical patterns behind voting decisions.

2. Project Objectives

1. Data Collection & Preprocessing: Ingest comprehensive NBA play-by-play data from Kaggle (via kagglehub API), clean data quality issues, standardize types, handle missing values, and partition by season for efficient querying in PySpark.
2. Developing PySpark Workflows: Implement scalable data transformation pipelines that extract 20+ features from natural language event descriptions using regex-based parsing and type-based event classification (eventmsgtype).
3. Feature Engineering at Scale: Derive advanced metrics including clutch performance indicators (4th quarter performance in tight games), efficiency ratios (FG%, FT%, True Shooting %), possession-based rates, and era-adjusted baselines from distributed event aggregations.
4. Scalable Analysis: Isolate 1.35M+ events into 1.35M clutch events for focused analysis; aggregate event-level data to player-season statistics; identify patterns in clutch vs. regular performance differentials; detect era bias in scoring trends.

5. MLlib Classification Modeling: Build machine learning models to predict official MVP vote shares from our 20+ engineered features, enabling data-driven MVP ranking independent of media narrative.
 6. Visualization & Performance Evaluation: Generate insights through statistical summaries, player rankings, efficiency vs. volume trade-off analysis, and evaluation of model performance on multiple cluster configurations.
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3. Data Description

Data Source:

- Kaggle NBA Dataset (via kagglehub Python API)
- Repository: wyattowalsh/basketball
- Access Method: `kagglehub.dataset_download("wyattowalsh/basketball")`

Data Size and Composition:

- Raw Data: 5 GB
- Total Play-by-Play Events: 1,350,000+ events
- Games Covered: 1,401 games across multiple seasons
- Temporal Coverage: 1990s through 2020s NBA seasons
- File Formats: CSV input, Parquet output (partitioned by `season_id`)

Data Characteristics:

- Structure: Highly structured relational data (three main CSV tables joined on IDs)
- Temporal: Time-series nature with timestamp, quarter, and game clock information per event
- Natural Language: Event descriptions contain unstructured text requiring regex parsing
- Scale: Massive event-level granularity (each possession = multiple events)

Data Challenges:

- Missing Values: Score margins with NULL/"TIE" entries; incomplete player names in some events
 - Unstructured Text: Natural language event descriptions require careful parsing (e.g., detecting "MISS 3PT" vs "MISS" vs "3PT")
 - High Dimensionality: 1.35M events × event fields → 20+ derived features per event
 - Imbalanced Time Distribution: More recent seasons have higher event density than 1990s data
 - Data Quality: Shooter name validation required; inconsistencies in event message formatting
 - Clutch Event Filtering: Complex temporal logic (4th quarter + ± 5 point margin + <5 minutes) reduces full dataset to 1.35M relevant clutch events
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4. Technical Approach

Core Big Data Stack:

- Apache Spark (PySpark): RDD operations for low-level transformations; DataFrames for SQL-based aggregations; MLlib for classification modeling
- Data Processing: Pandas (local aggregation), NumPy (numerical operations)
- Visualization: Matplotlib, Seaborn (static analysis plots)
- ML & Evaluation: scikit-learn (model evaluation metrics), MLlib (distributed classification)

Cluster Usage Plan:

- Nodes: 1-4 CPU nodes (scaling evaluation)
- Job Type: Batch processing with iterative ML training cycle
- Storage Format: Parquet

- Partition Strategy: By season_id for seasonal analysis

Processing Stages:

- Stage 1 (Extract/Ingest): Sequential download and schema inference (single-machine bottleneck)
- Stage 2 (Transform): Distributed data cleaning, type casting, null handling (scales with node count)
- Stage 3 (Feature Engineering): Regex-based parsing, event classification across partitions (high parallelism)
- Stage 4 (Aggregation): Player-season summarization via groupBy operations (shuffle-intensive)

Pipeline Overview:

1. Data Ingestion and Preprocessing

- Download Kaggle NBA dataset via API
- Parse CSV files into PySpark DataFrames
- Schema validation and type inference
- Join play_by_play with game metadata and player info
- Handle missing values (NULL score margins → 0; "TIE" → 0)
- Output cleaned data as Parquet (partitioned by season_id)

2. Exploratory Analysis

- DataFrame row/column counts; data type verification
- Summary statistics (min/max/mean for numeric fields)
- Null value analysis by column
- Temporal distribution of events (events per season, per game)
- Player coverage (unique player IDs, appearances)

3. Feature Engineering

- Parse event descriptions via F.instr() for MISS, 3PT, rebound, steal, block, assist

- Classify events using eventmsgtype (1=FG make, 2=FG miss, 3=FT, 5=turnover, 6=foul)
- Compute points per event: 3 (3P make) + 2 (FG make) + 1 (FT make)
- Calculate shooting splits: $FG\% = fg_make / (fg_make + fg_miss)$, FT%, 3P%
- Define clutch events: $period==4$ AND $|score_margin|\leq 5$ AND $seconds_remaining\leq 300$
- Aggregate clutch events to player-season totals
- Derive possession-based metrics: usage rate, points_per_possession, turnover frequency
- Compute era-adjusted baselines (clutch points/game per decade)

4. Model Building and Evaluation

- Consolidate 20+ features into player-season feature vectors
- Prepare labels: official MVP vote share percentages
- Feature importance analysis to identify MVP drivers
- Prediction on held-out test seasons

5. Visualization and Interpretation of Results

- Top clutch scorers (LeBron, Kobe, KD rankings)
- Efficiency leaders (FG% rankings)
- Era bias trends (clutch points/game across decades)
- Efficiency vs. volume scatter plots

5. Results and Discussion

Volume vs. Efficiency Divergence:

- LeBron James leads all-time clutch scorers with 1,509 points across 1,192 games (PPP=0.438)
- Andris Biedrins dominates clutch FG% with 48.7% despite only 76 attempts

- This divergence shows that raw volume does not correlate with shooting efficiency, where high-usage scorers sacrifice accuracy for opportunity volume

Era Bias Pattern:

- 1990s clutch scoring: 5.05 points/game baseline
- 2020s clutch scoring: 5.63 points/game baseline
- Modern eras show 11% higher clutch scoring per game, likely driven by rule changes (defensive three-second elimination), improved spacing, and increased pace

Underrated vs. Overrated Clustering:

- High-usage, low-efficiency players ($3+ \text{ possessions/game}$, $\text{PPP} \leq 0.9$): inflate clutch narrative despite poor conversion
- Low-usage, high-efficiency players ($\leq 2 \text{ possessions/game}$, $\text{PPP} \geq 1.1$): role players whose impact is severely undervalued by MVP voting
- This finding suggests MVP Index must normalize for the usage/efficiency trade-off

Features Ranked by Importance:

- 1. Points per possession (highest impact on MVP vote prediction)
- 2. Clutch FG% (shot selection under pressure)
- 3. Era-adjusted baseline (accounting for decade effects)
- 4. Usage rate (volume component)
- Assist rate (playmaking impact)

6. Credit Statement

Siddarth Gupta: Data Curation, Software, Formal Analysis

- Aggregated cleaned clutch events at the player level; computed totals and efficiency metrics; saved final player-level dataset for modeling.

Anoop Ibrampur: Software, Methodology, Conceptualization

- Developed the full Spark-based feature engineering pipeline; extracted per-player season metrics including scoring, usage, clutch performance, and designed the weighted Impact Score formula.

Abdulrahman Osilan: Software, Data Curation, Validation

- Executed clutch event engineering pipeline; applied filtering logic to isolate clutch plays; validated data integrity and timestamps.

Vansh Koul: Formal Analysis, Visualization

- Conducted final analytics and visualizations; produced clutch player rankings, efficiency analysis, era bias evaluation; created presentation plots and summary tables.

Ryan Le: Writing - Original Draft, Writing - Review & Editing, Supervision, Project Administration

- Led presentation creation and final report composition; coordinated team communication and meetings; refined coding approaches collaboratively.

7. References

<https://www.nba.com/stats>

<https://www.basketball-reference.com>

<https://www.espn.com/nba/stats>

<https://www.kaggle.com/datasets/wyattowalsh/basketball>
